

CIS270A Project 2 Requirements

GTC Telephone Company wants a program to update its customer master file (GTCSTP) based on data contained in the payments transaction file (GTCPAYP).

GTCSTP: Customer Master File

This file contains a record for each of GTC's customers.

*This "Master" file will be **RANDOMLY** accessed by using the **Chain operation**, and will be updated; you will make your own copy of this file called **MYCSTP***

Record Layout of GTCSTP

Record Format CUSREC

Field	Description	Length/Type	(Decimal positions)	Notes
*CPHONE	Customer phone number	10S	(0)	UNIQUE
CLNAME	Last name	15	--	--
CFNAME	First name	10	--	--
CSTRET	Street address	20	--	--
CCITY	City	15	--	--
CSTAT	State	2	--	--
CZIP	Postal code	5S	(0)	--
CURBIL	Current billing amount	6S	(2)	--
AMTOWE	Amount owed	6S	(2)	--
PAYDATL	Date last payment	L	--	Native date
PAYDAT	Date last payment	8S	(0)	ccyyymmdd

* denotes key

GTCPAYP: Payments Transaction File

File GTPAYP is generated through optical character recognition (OCR) and manual-entry techniques. Records are used once to update the customer account figures and generate a payment report, and are then archived.

*This "Payment" input file will be **SEQUENTIALLY** accessed by using the **Read operation**.*

Record Format PAYREC

Field	Description	Length/Type	(Decimal positions)	Notes
CPHONE	Payer's phone number	10S	(0)	--
AMTPD	Amount paid	6S	(2)	--
DATRCVL	Date payment received	L		Native date
DATRCV	Date payment received	8S	(0)	ccyyymmdd

REPORT1 / TOTALLINE										  																													
MM/DD/YY										GTC Telephone Company										Page 0001																			
										Payment Processing Transaction Audit Report																													
Customer										Date Received										Amount																			
6666666666										YYYY-MM-DD										666666										*** Customer Record Not Found ***									
																				666										*** Customers not found in Master file ***									

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BEFORE YOU START:

It will be necessary for you to have your “own” copy of the GTCSTP file, so that you are NOT updating the file in the RPGIV_5THS library, which is on everyone’s library list. Instead, you will be updating your OWN copy of this file.

To avoid possibly confusing “your” file with the “real” file, we will have it renamed during the copy process:

```
CPYF FROMFILE(RPGIV_5THS/GTCSTP) TOFILE(yourlibrary/MYCSTP) CRTFILE(*YES)
```

Therefore, in your programming, refer to MYCSTP for the file name instead of GTCSTP.

Every time you run your program, it will update the file! Therefore, whenever you need to “start fresh” with the original file contents:

```
DLTF YOURLIB/MYCSTP
```

```
CPYF FROMFILE(RPGIV_5THS/GTCSTP) TOFILE(yourlibrary/MYCSTP) CRTFILE(*YES)
```

Familiarize yourself with the contents of both disk files by using the RUNQRY command and by downloading the data into Excel spreadsheets for reference. Both files are found in the RPGIV_5THS library. Write down or highlight some of the customer numbers, dates, amounts so that you can verify that your program properly updates the data.

```
RUNQRY QRYFILE((diskFileName))
```

Program Notes:

1. Define the audit report using RDi Report Designer to generate the DDS printer definition, following the printchart above.
 - Add your name to the top of the report.
 - An audit report lists the changes applied to the master file, therefore, the customer number, date received and amount on the audit report are from the GTCPAYP file.
 - Even though this printer spacing chart does not show editing of the amount, it should be edited for readability.
 - If a customer in the payment file is not found in the customer file, the report is to display an error notation; a count of invalid records should be displayed at the end of report.
2. Code the RPG program to sequentially process the GTCPAYP payment file. For each record in the payment file, randomly retrieve the appropriate record from the MYCSTP customer master file, and subtract the payment amount from the amount owed. Update the customer master record with the updated amount owed and the payment date. The use of UPDATE with %FIELDS in your solution is required, and is also the most efficient approach. Be sure to include commenting as outlined in Project 1.
 - AMTOWE – should reflect the new amount owed
 - PAYDATL – should contain the native date *from the payment transaction file*
 - PAYDAT – should contain the legacy date from the payment transaction file
3. Check your MYCSTP data file. The data file MYCSTP should show updated data – the date of last payment fields (both of them) and amount owed should be updated. Remember to refresh your copy of this file EVERY time you run the program.
4. Do your housekeeping: leaving only the last good compilations of the DDS and RPG program and the output report itself, housekeeping is part of your grade. Email me when your reports and updated file are ready to be evaluated. I encourage you to continue to utilize the RDi software.